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A multiaxial perfectly matched layer (M-PML) for the long-time simulation of elastic wave propagation in the second-order equations

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ABSTRACT

In order to conquer the spurious reflections from the truncated edges and maintain the stability in the long-time simulation of elastic wave propagation, several perfectly matched layer (PML) methods have been proposed in the first-order (e.g., velocity–stress equations) and the second-order (e.g., energy equation with displacement unknown only) formulations. The multiaxial perfectly matched layer (M-PML) holds the excellent stability for the long-time simulation of wave propagation, even though it is not perfectly matched in the discretized M-PML equation system. This absorbing boundary approach can offer an alternative way to solve the problem of the late-time instability, especially for anisotropic media, which is also suffered by the convolutional perfectly matched layer (C-PML) that is supposed to be competent to handle most stable problems. The M-PML termination implementation in the first-order formulations is well proposed. The common drawback of the implementation of the first-order M-PML formulations is that it necessitates fundamental reconstruction of the existing codes of the second-order spectral element method (SEM) or finite element method (FEM). Therefore, we propose a nonconvolutional second-order M-PML absorbing boundary condition approach for the wave propagation simulation in elastic media that has not yet been developed before. Two-dimensional numerical simulation validations demonstrate that the proposed second-order M-PML has good performances: 1) superior efficiency and stability of absorbing the spurious elastic wavefields, both the surface waves and body waves, reflected on the boundaries; 2) superior stability in the long-time simulation even in the isotropic medium with a high Poisson's ratio; 3) superior efficiency and stability in the long-time simulation for anisotropic media. This method hence makes the SEM and FEM in the second-order wave equation formulation more efficient and stable for the long-time simulation.

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1. Introduction

A number of absorbing boundary conditions (ABCs) have been developed in wave propagation modeling in recent decades, which are necessary techniques to suppress spurious reflections from the truncated edges of a finite-sized discrete earth model. The classical ABCs, 'sponge zones' (Cerjan et al., 1985; Sochacki et al., 1987) and paraxial boundaries (Clayton and Engquist, 1977; Engquist and Majda, 1977; Higdon, 1991; Quarteroni et al., 1998; Stacey, 1988), etc., have been proposed and demonstrated great efficiency in simulating seismic wave propagation. However, the angle and the frequency of incident waves, the elastic properties of the earth model as well, obviously affect the absorbing efficiency.

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The perfectly matched layer (PML), introduced by Bérenger (1994), was firstly used in Maxwell equations. In order to make it possible to incorporate mathematical formulations for desired absorbency, the field variables of the PML are split into nonphysical components. Because of the advantages of independences on the angle and the frequency of incident waves, researchers promote the PML method in different wave propagation modeling problems, as electromagnetic wave propagation (e.g., Bérenger, 1996, 1998; Collino and Monk, 1998; Harari et al., 2000; Teixeira and Chew, 1998a, b), acoustical wave propagation (e.g., Hesthaven, 1998; Hu, 1996; Nataf, 2006), as well as elastic wave propagation in elastic media (e.g., Basu and Chopra, 2004; Chew and Liu, 1996; Collino and Tsogka, 2001; Festa and Nielsen, 2003; Hastings et al., 1996), in viscoelastic media (Bécache et al., 2004; Martin and Komatitsch, 2009), and in poroelastic media (Martin et al., 2008; Zeng et al., 2001).

Although the numerical validations of this method are successful, the PML still faces many challenges, e.g., poor absorbency for strong reflections at grazing incidences, and instability for the long-time simulation (Festa et al., 2005; Komatitsch and Martin, 2007; Matzen, 2011). These problems have been intensively investigated in recent years (Halpern et al., 2011). For example, due to the divergence of the PML solution under small perturbations (Abarbanel and Gottlieb, 1997), it was initially thought that noncausality of the constitutive parameters makes dynamical instability of the classical PML (Kuzuoglu and Mittra, 1996). The convolutional perfectly matched layer (C-PML), an alternative causal frequency dependent PML, was hence proposed and particularly investigated (Abarbanel and Gottlieb, 1998; Appelö and Kreiss, 2005; Basu and Chopra, 2004; Gedney, 1996; Komatitsch and Martin, 2007; Li and Matar, 2010; Matzen, 2011; Zhao and Cangellaris, 1996; Ziolkowski, 1997). The absorbency of the reflected waves from incident outgoing waves on artificial boundaries at grazing angle is improved (Bérenger, 2002; Festa and Vilotte, 2005; Komatitsch and Martin, 2007; Martin and Komatitsch, 2009; Martin et al., 2008; Roden and Gedney, 2000). However, the C-PML cannot solve all the instable problems (Komatitsch and Martin, 2007). It is believed to suffer from the late-time instability, especially for anisotropic media (Matzen, 2011). Meanwhile, the C-PML loses its absorptive capacity in an elastic medium at low frequencies (Meza-Fajardo and Papageogiou, 2008). The multiaxial PML (M-PML) method, which is efficient and stable without dependence on the frequency and the direction of wave propagation, was hence developed (Meza-Fajardo and Papageogiou, 2008, 2010, 2012). The M-PML method is based upon a more general coordinate stretching version of the classical split-field PML, in which the waves are absorbed in all directions with different damping factors. Zeng et al. (2011) validated the excellent stability of the M-PML method in the media with high Poisson's ratios. Due to effective absorption of the spurious reflections, even though it is not perfectly matched in the discretized M-PML formulation system (Meza-Fajardo and Papageogiou, 2012), the M-PML is implemented for anisotropic elastic media with good performance (Dmitriev and Lisitsa, 2011, 2012). The M-PML method has been verified to be more stable and efficient than the classical PML and the C-PML in the long-time simulation of wave propagation in isotropic and anisotropic media in the first-order equation formulation (Meza-Fajardo and Papageogiou, 2008). Meza-Fajardo and Papageogiou (2012) further pointed out that increasing the layer thickness has very important impact on (i.e., is more effective in) reducing reflections. Through their numerical examples, it illustrated that the overall reflections of the M-PML are the same or less than that obtained when the classical PML and the C-PML are used.

For the second-order wave equations, however, the first-order M-PML cannot be applied straightforwardly. The common drawback of implementation of these formulations is that it necessitates fundamental reconstruction of the existing codes of the finite element method (FEM) or spectral element method (SEM) in the second-order equations. In addition, the reconstructed codes of the FEM or SEM may in common consume more computing resources, due to system matrices with extra non-zero entries (and broken symmetries) implying growth in storage, and the number of flops to be performed at each time step, since, apart from displacements, stresses additionally become the unknowns. Furthermore, most of the existing FEM software for solid mechanical problems solves the wave equations for the displacements only. Therefore, it is an urgent need that the PML formulations are established for the second-order elastic wave equations. Komatitsch and Tromp (2003) firstly introduced a classical PML for the second-order wave equations in elastic media. Their PML is highly absorbent for the surface waves in elastic media, but unstable in the long-time simulation. Li and Matar (2010) proposed a non-split C-PML for the second-order mixed wave equations in elastic media. In reality, their scheme formulations are also needed in reconstructing the existing second-order FEM or SEM codes. Matzen (2011) explicitly investigated non-split C-PML formulations for the second-order wave equations in elastic media. He believed that the second-order C-PML would suffer from the late-time instabilities in anisotropic media. To our best knowledge, the PML approaches to conquer the late-time instabilities for wave simulation in the second-order equations, especially for anisotropic media, are rather limited. Whereas the M-PML has offered an applicable way to solve the problem in the first-order equations (Meza-Fajardo and Papageogiou, 2008, 2012; Zeng et al., 2011), however, the applications of the M-PML in the context of the second-order wave equations are not explicitly studied. Thus, the promotion of the M-PML into the second-order equations has still to be suggested to meet the urgent need.

2. The classical second-order PML formulations

The elastic wave equation can be written as

$$\rho\partial_t{}^2\mathbf{s} = \nabla\cdot\left[\mathbf{c}:\frac{1}{2}\left(\nabla\mathbf{s}+\nabla^T\mathbf{s}\right)\right], \tag{1}$$

where $\mathbf{s}$ is the unknown displacement vector, $\mathbf{c}$ is the elastic tensor and ρ is the mass density. Eq. (1) also has the form in the frequency-domain of

$$-\rho\omega^2\mathbf{s}=\nabla\cdot\left[\mathbf{c}:\frac{1}{2}\left(\nabla\mathbf{s}+\nabla^T\mathbf{s}\right)\right], \tag{2}$$

where ω is the angular frequency. In order to obtain the PML formulations of Eq. (1), the gradient operator ∇ can be split into the components perpendicular and parallel to the boundary between the regular and the PML region.

$$\boldsymbol{\nabla} = \mathbf{n}\partial_n + \boldsymbol{\nabla}^{//}, \tag{3}$$

where the local normal direction of the boundary between the regular and the PML region is denoted by the vector $\mathbf{n}$ in Fig. 1 (always indicates the normal vector of the coordinates). And $\boldsymbol{\nabla}^{//} = (\mathbf{I}-\mathbf{nn})\cdot\boldsymbol{\nabla} = \mathbf{m}\partial_m$, the parallel direction of the boundary is presented as the vector $\mathbf{m}$ in Fig. 1. Damping of the wave equation can be accomplished by introducing a new variable

$$\tilde{n}(n) = n-\frac{i}{\omega}\int_0^n d(\zeta)d\zeta. \tag{4}$$

Therefore, the relation between the new and the regular variable can be expressed by

$$\begin{aligned}&\frac{\partial n}{\partial\tilde{n}} = \frac{i\omega}{i\omega+d(n)},\\ &\frac{\partial}{\partial n}\left(\frac{\partial n}{\partial\tilde{n}}\right) = -\frac{i\omega}{(i\omega+d(n))^2}\frac{\partial d(n)}{\partial n},\end{aligned} \tag{5}$$

where $d(n)$, a function of the distance to the interface n, is a damping profile across the PML region. Then, if we substitute Eq. (3) into Eq. (2), and replace the wave equation written in n with the generalized variable $\tilde{n}$ mentioned in Eq. (4), we can obtain

$$\nabla = \mathbf{n}\partial_{\tilde{n}} + \nabla^{//}. \tag{6}$$

The PML formulations can be rewritten as

$$\begin{aligned} -\rho\omega^2\mathbf{s} = {} & \mathbf{n}\partial_{\tilde{n}}\cdot\left[\mathbf{c}:\frac{\mathbf{1}}{2}\left(\left(\mathbf{n}\partial_{\tilde{n}}\mathbf{s}\right)+\left(\mathbf{n}\partial_{\tilde{n}}\mathbf{s}\right)^T\right)\right] + \mathbf{n}\partial_{\tilde{n}}\cdot\left[\mathbf{c}:\frac{\mathbf{1}}{2}\left(\left(\nabla^{//}\mathbf{s}\right)+\left(\nabla^{//}\mathbf{s}\right)^T\right)\right] \\ & +\nabla^{//}\cdot\left[\mathbf{c}:\frac{\mathbf{1}}{2}\left(\left(\mathbf{n}\partial_{\tilde{n}}\mathbf{s}\right)+\left(\mathbf{n}\partial_{\tilde{n}}\mathbf{s}\right)^T\right)\right] + \nabla^{//}\cdot\left[\mathbf{c}:\frac{\mathbf{1}}{2}\left(\left(\nabla^{//}\mathbf{s}\right)+\left(\nabla^{//}\mathbf{s}\right)^T\right)\right]. \end{aligned} \tag{7}$$

The summation of the split displacements represents the real displacement in the PML region.

$$\mathbf{s} = \mathbf{s}^{\mathbf{1}} + \mathbf{s}^{2} + \mathbf{s}^{3} + \mathbf{s}^{4}, \tag{8}$$

where $-\rho\omega^2\mathbf{s}^{\mathbf{1}} \sim -\rho\omega^2\mathbf{s}^{4}$ denote the four components of the right part of Eq. (7). Then, the weak forms of the split PML formulations in the time domain can be rewritten by Komatitsch and Tromp (2003)

$$\begin{aligned} \int_V \rho(\partial_t + d)^2\mathbf{s}^{\mathbf{1}}\cdot\mathbf{w}d^3\mathbf{r} = {} & -\int_V (\mathbf{n}\partial_n\mathbf{w}):\mathbf{c}:\frac{1}{2}\left((\mathbf{n}\partial_n\mathbf{s})+(\mathbf{n}\partial_n\mathbf{s})^T\right)d^3\mathbf{r} \\ & +\int_\Gamma (\mathbf{nw}):\mathbf{c}:\frac{1}{2}\left((\mathbf{n}\partial_n\mathbf{s})+(\mathbf{n}\partial_n\mathbf{s})^T\right)d^2\mathbf{r}, \end{aligned}$$

$$\int_V \rho(\partial_t + d)^3\mathbf{s}^{\mathbf{2}}\cdot\mathbf{w}d^3\mathbf{r} = -\int_V d'\mathbf{nw}:\mathbf{c}:\frac{1}{2}\left((\mathbf{n}\partial_n\mathbf{s})+(\mathbf{n}\partial_n\mathbf{s})^T\right)d^3\mathbf{r},$$

$$\begin{aligned} \int_V \rho\partial_t(\partial_t + d)\mathbf{s}^{\mathbf{3}}\cdot\mathbf{w}d^3\mathbf{r} = {} & -\int_V \left[(\mathbf{n}\partial_n\mathbf{w}):\mathbf{c}:\frac{1}{2}\left(\left(\nabla^{//}\mathbf{s}\right)+\left(\nabla^{//}\mathbf{s}\right)^T\right)\right. \\ & \left.+\left(\nabla^{//}\mathbf{w}\right):\mathbf{c}:\frac{1}{2}\left((\mathbf{n}\partial_n\mathbf{s})+(\mathbf{n}\partial_n\mathbf{s})^T\right)\right]d^3\mathbf{r} + \int_\Gamma (\mathbf{nw}):\mathbf{c}:\frac{1}{2}\left(\left(\nabla^{//}\mathbf{s}\right)+\left(\nabla^{//}\mathbf{s}\right)^T\right)d^2\mathbf{r}, \end{aligned}$$

$$\int_V \rho\partial_t^{\,2}\mathbf{s}^{\mathbf{4}}\cdot\mathbf{w}d^3\mathbf{r} = -\int_V \left(\nabla^{//}\mathbf{w}\right):\mathbf{c}:\frac{1}{2}\left(\left(\nabla^{//}\mathbf{s}\right)+\left(\nabla^{//}\mathbf{s}\right)^T\right)d^3\mathbf{r}, \tag{9}$$

where $\mathbf{w}$ represents the test function, Γ denotes the boundary of the PML that does not contact the regular region.

3. The M-PML formulations

In the classical PML, there is only one damping profile, $d(n)$, which is specified to be a function of the distance to the interface n, while the other two profiles, e.g. $d(m)$ and $d(l)$, whose directions are perpendicular to $\mathbf{n}$, are equal to 0, i.e.

$$d_n = d_n(n), d_m = 0, d_l = 0. \tag{10}$$

Nevertheless, the damping profiles in the proposed M-PML can be rewritten as the functions of the n variable

$$d_n = d_n^{(n)}(n), d_m = d_m^{(n)}(n), d_l = d_l^{(n)}(n). \tag{11}$$

The additional damping profiles $d_m^{(n)}(n)$ and $d_l^{(n)}(n)$ are proportional to $d_n^{(n)}(n)$ (Meza-Fajardo and Papageogiou, 2008):

$$d_m^{(n)}(n) = p^{(m/n)}d_n^{(n)}(n), d_l^{(n)}(n) = p^{(l/n)}d_n^{(n)}(n), \tag{12}$$

where $p^{(m/n)}$ and $p^{(l/n)}$ are the constants for fine-tuning of the stability of the M-PML region.

The design of the M-PML is achieved by interpreting it not as a medium of complex space variables, but as a medium with complex wave vectors.

$$\tilde{k}_n = k_n + \frac{1}{i\omega}\alpha_n^{(n)}k_n, \quad \tilde{k}_m = k_m + \frac{1}{i\omega}\alpha_m^{(n)}k_m, \quad \tilde{k}_l = k_l + \frac{1}{i\omega}\alpha_l^{(n)}k_l, \tag{13}$$

where $\alpha_n^{(n)}, \alpha_m^{(n)}$ and $\alpha_l^{(n)}$ are the functions related to the damping profiles $d_n^{(n)}$, $d_m^{(n)}$ and $d_l^{(n)}$, which can be written as

$$\alpha_n^{(n)} = \frac{1}{n}\int_0^n d_n^{(n)}(\zeta)d\zeta, \alpha_m^{(n)} = d_m^{(n)}(n), \alpha_l^{(n)} = d_l^{(n)}(n). \tag{14}$$

In order to investigate the efficiency of the M-PML formulations, the plane-wave solution can be expressed as

$$\mathbf{u} = \mathbf{A}\exp[-i(\mathbf{k}\cdot\mathbf{r}-\omega t)], \tag{15}$$

where $\mathbf{A}$ is the polarization vector, $\mathbf{k}$ is the wave vector and $\mathbf{r}$ is the position vector.

Substituting the transformations of Eq. (13) into Eq. (15), the solution of Eq. (15) can be rewritten as

$$\mathbf{u} = \mathbf{A}\exp(-\boldsymbol{\alpha}\cdot\mathbf{r})\exp[-i(\mathbf{k}\cdot\mathbf{r}-\omega t)], \boldsymbol{\alpha} = \frac{|\mathbf{k}|}{\omega}\left[\begin{array}{ccc} e_n\alpha_n^{(n)} & e_m\alpha_m^{(n)} & e_l\alpha_l^{(n)} \end{array}\right], \tag{16}$$

where e_n, e_m, and e_l are the direction cosines of the wave vector $\mathbf{k}$. The factor $\exp(-\boldsymbol{\alpha}\cdot\mathbf{r})$ implies the wavefileds in the M-PML region decaying in the direction of the wave vector.

The transformations that result from the application of the variable stretching approach in the M-PML formulations are written as

$$\tilde{n} = n + \frac{1}{i\omega}\alpha_n^{(n)}n, \quad \tilde{m} = m + \frac{1}{i\omega}\alpha_m^{(n)}m, \quad \tilde{l} = l + \frac{1}{i\omega}\alpha_l^{(n)}l. \tag{17}$$

Comparing Eq. (17) with Eq. (4), we find that they have the same form. Then, the M-PML may be accomplished by introducing new variables $\tilde{n}$, $\tilde{m}$ and $\tilde{l}$ in Eq. (17).

The corresponding partial derivatives of the stretching variables are written by (Meza-Fajardo and Papageogiou, 2008)

$$\frac{\partial\tilde{n}}{\partial n} = 1 + \frac{1}{i\omega}d_n^{(n)}, \quad \frac{\partial\tilde{m}}{\partial m} = 1 + \frac{1}{i\omega}d_m^{(n)}, \quad \frac{\partial\tilde{l}}{\partial l} = 1 + \frac{1}{i\omega}d_l^{(n)}. \tag{18}$$

4. The second-order M-PML formulations

For two-dimensional case, we suppose that the boundary between the regular region and the PML region is a line with the normal direction represented by $\mathbf{n}$, and the parallel direction represented by $\mathbf{m}$ (see Fig. 1).

In the classical second-order PML system, Eq. (6) represents the waveform damping along the direction $\mathbf{n}$. There is no damping in the direction parallel to the boundary for the parallel gradient $\nabla^{//}$. In order to make the wavefields decay in all the directions of the wave vector $\mathbf{k}$, the parallel gradient $\nabla^{//}$ needs to be operated in the PML. The operate $\nabla^{//}$ can be rewritten as

$$\nabla^{//} = \mathbf{m}\partial_m, \tag{19}$$

where $\mathbf{m}$ is parallel to the boundary.

The M-PML system can be accomplished by replacing both n and m with the generalized variable $\tilde{n}$ and $\tilde{m}$ in Eq. (19), respectively. Therefore,

$$\nabla^{//} = \mathbf{m}\partial_{\tilde{m}}. \tag{20}$$

The new complex variable $\tilde{m}$ in the M-PML has the form as

$$\tilde{m} = m + \frac{1}{i\omega}\alpha_m{}^{(n)} m, \tag{21}$$

where the function $\alpha_m{}^{(n)} = p^{(m/n)} d(n)$ and the constant $p^{(m/n)}$ are also for fine-tuning the stability of the M-PML region. Therefore, the relation between the new and the regular variables can be also expressed by

$$\frac{\partial m}{\partial \tilde{m}} = \frac{i\omega}{i\omega + d(m)}, \tag{22}$$

where $d(m) = p^{(m/n)} d(n)$. Then,

$$\begin{aligned} \frac{\partial}{\partial n}\left(\frac{\partial m}{\partial \tilde{m}}\right) &= -\frac{i\omega}{(i\omega + d(m))^2} p^{(m/n)} d'(n), \\ \frac{\partial}{\partial m}\left(\frac{\partial m}{\partial \tilde{m}}\right) &= 0, \\ \frac{\partial}{\partial m}\left(\frac{\partial n}{\partial \tilde{n}}\right) &= 0, \end{aligned} \tag{23}$$

Substituting Eq. (20) into Eq. (6), the gradient operator ∇ can be expressed as

$$\nabla = \mathbf{n}\partial_{\tilde{n}} + \mathbf{m}\partial_{\tilde{m}}. \tag{24}$$

Therefore, by substituting Eq. (24) into Eq. (2), the wave equation in the frequency domain can be rewritten as

$$\begin{aligned} -\rho\omega^2 \mathbf{s} = {} & \mathbf{n}\partial_n \cdot \left[\mathbf{c}:\frac{1}{2}\left((\mathbf{n}\partial_n \mathbf{s}) + (\mathbf{n}\partial_n \mathbf{s})^T\right)\right](\partial n/\partial\tilde{n})^2 \\ & + \mathbf{n} \cdot \left[\mathbf{c}:\frac{1}{2}\left((\mathbf{n}\partial_n \mathbf{s}) + (\mathbf{n}\partial_n \mathbf{s})^T\right)\right](\partial n/\partial\tilde{n})\partial_n(\partial n/\partial\tilde{n}) \\ & + \left\{\mathbf{n}\partial_n \cdot \left[\mathbf{c}:\frac{1}{2}\left(\left(\nabla^{//}\mathbf{s}\right) + \left(\nabla^{//}\mathbf{s}\right)^T\right)\right]\right. \\ & \left. + \nabla^{//} \cdot \left[\mathbf{c}:\frac{1}{2}\left((\mathbf{n}\partial_n \mathbf{s}) + (\mathbf{n}\partial_n \mathbf{s})^T\right)\right]\right\}(\partial n/\partial\tilde{n})(\partial m/\partial\tilde{m}) \\ & + \mathbf{n} \cdot \left[\mathbf{c}:\frac{1}{2}\left(\left(\nabla^{//}\mathbf{s}\right) + \left(\nabla^{//}\mathbf{s}\right)^T\right)\right](\partial n/\partial\tilde{n})\partial_n(\partial m/\partial\tilde{m}) \\ & + \mathbf{m} \cdot \left[\mathbf{c}:\frac{1}{2}\left((\mathbf{n}\partial_n \mathbf{s}) + (\mathbf{n}\partial_n \mathbf{s})^T\right)\right](\partial m/\partial\tilde{m})\partial_m(\partial n/\partial\tilde{n}) \\ & + \nabla^{//} \cdot \left[\mathbf{c}:\frac{1}{2}\left(\left(\nabla^{//}\mathbf{s}\right) + \left(\nabla^{//}\mathbf{s}\right)^T\right)\right](\partial m/\partial\tilde{m})^2 \\ & + \mathbf{m} \cdot \left[\mathbf{c}:\frac{1}{2}\left(\left(\nabla^{//}\mathbf{s}\right) + \left(\nabla^{//}\mathbf{s}\right)^T\right)\right](\partial m/\partial\tilde{m})\partial_m(\partial m/\partial\tilde{m}). \end{aligned} \tag{25}$$

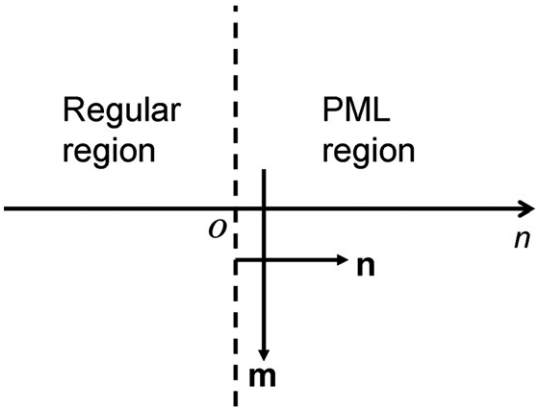


Fig. 1. Relation of the regular region and the PML region. **n** and **m** denote the normal and the parallel directions of the interface, respectively. They all indicate the damping directions.

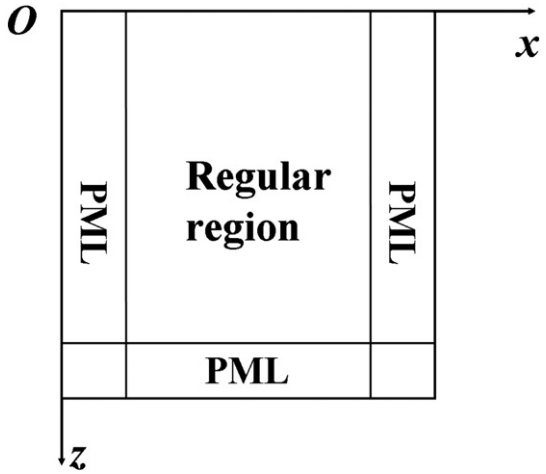


Fig. 2. The geometry of a PML model. The PML boundaries are set in vertical and bottom edges.

Then, we can split the displacement vector into five parts,

$$\mathbf{s} = \mathbf{s}^1 + \mathbf{s}^2 + \mathbf{s}^3 + \mathbf{s}^4 + \mathbf{s}^5. \tag{26}$$

Eq. (25) can be rewritten as

$$\begin{aligned} -\rho\omega^2 \mathbf{s}^1 &= \mathbf{n}\partial_n \cdot \left[\mathbf{c}:\frac{1}{2}\left((\mathbf{n}\partial_n \mathbf{s}) + (\mathbf{n}\partial_n \mathbf{s})^T\right)\right](\partial n/\partial\tilde{n})^2, \\ -\rho\omega^2 \mathbf{s}^2 &= \mathbf{n} \cdot \left[\mathbf{c}:\frac{1}{2}\left((\mathbf{n}\partial_n \mathbf{s}) + (\mathbf{n}\partial_n \mathbf{s})^T\right)\right](\partial n/\partial\tilde{n})\partial_n(\partial n/\partial\tilde{n}), \\ -\rho\omega^2 \mathbf{s}^3 &= \left\{\mathbf{n}\partial_n \cdot \left[\mathbf{c}:\frac{1}{2}\left(\left(\nabla^{//}\mathbf{s}\right) + \left(\nabla^{//}\mathbf{s}\right)^T\right)\right]\right. \\ &\quad \left. + \nabla^{//} \cdot \left[\mathbf{c}:\frac{1}{2}\left((\mathbf{n}\partial_n \mathbf{s}) + (\mathbf{n}\partial_n \mathbf{s})^T\right)\right]\right\}(\partial n/\partial\tilde{n})(\partial m/\partial\tilde{m}), \\ -\rho\omega^2 \mathbf{s}^4 &= \nabla^{//} \cdot \left[\mathbf{c}:\frac{1}{2}\left(\left(\nabla^{//}\mathbf{s}\right) + \left(\nabla^{//}\mathbf{s}\right)^T\right)\right](\partial m/\partial\tilde{m})^2, \\ -\rho\omega^2 \mathbf{s}^5 &= \mathbf{n} \cdot \left[\mathbf{c}:\frac{1}{2}\left(\left(\nabla^{//}\mathbf{s}\right) + \left(\nabla^{//}\mathbf{s}\right)^T\right)\right](\partial n/\partial\tilde{n})\partial_n(\partial m/\partial\tilde{m}). \end{aligned} \tag{27}$$

The displacement vector in the second-order formulations becomes to be represented by five components in the M-PML region, one more than those for the classical PML method. The weak form of the second-order M-PML formulations is obtained by multiplying Eq. (27) with the test vector **w** and converting them back into the time domain as

Table 1
Parameters of isotropic models.

Material properties	Case 1[a]	Case 2[b]
P wave velocity $V_P(m/s)$	2000	800
S wave velocity $V_S(m/s)$	1154.7	200
Mass density $\rho(kg/m^3)$	2200	2000
Poisson's ratio ν	0.25	0.47

[a] Komatitsch and Tromp (2003).
[b] Zeng et al. (2011).

Table 2
Parameters of anisotropic models.

Material properties	Case 3[a]	Case 4[b]
Mass density $\rho(kg/m^3)$	5000	7100
C_{11} (N/m^2)	2.00×10^{10}	1.65×10^{11}
C_{12} (N/m^2)	3.75×10^{10}	5.00×10^{10}
C_{22} (N/m^2)	1.00×10^{11}	6.20×10^{10}
C_{33} (N/m^2)	1.00×10^{10}	3.96×10^{10}

[a] Bécache et al. (2001).
[b] Meza-Fajardo and Papageogiou (2008).

$$\int_V \rho(\partial_t + d(n))^2 \mathbf{s^1}\cdot\mathbf{w}d^2\mathbf{r} = -\int_V (\mathbf{n}\partial_n\mathbf{w}):\mathbf{c}:\frac{1}{2}\left((\mathbf{n}\partial_n\mathbf{s})+(\mathbf{n}\partial_n\mathbf{s})^T\right)d^2\mathbf{r} + \int_\Gamma (\mathbf{nw}):\mathbf{c}:\frac{1}{2}\left((\mathbf{n}\partial_n\mathbf{s})+(\mathbf{n}\partial_n\mathbf{s})^T\right)d\mathbf{r},$$

$$\int_V \rho(\partial_t + d(n))^3 \mathbf{s^2}\cdot\mathbf{w}d^2\mathbf{r} = -\int_V d'(n)\mathbf{nw}:\mathbf{c}:\frac{1}{2}\left((\mathbf{n}\partial_n\mathbf{s})+(\mathbf{n}\partial_n\mathbf{s})^T\right)d^2\mathbf{r},$$

$$\int_V \rho\left[\partial_t{}^2 + (d(m)+d(n))\partial_t + d(m)d(n)\right]\mathbf{s^3}\cdot\mathbf{w}d^2\mathbf{r} = -\int_V [(\mathbf{n}\partial_n\mathbf{w}):\mathbf{c}:\frac{1}{2}\left(\left(\nabla^{//}\mathbf{s}\right)+\left(\nabla^{//}\mathbf{s}\right)^T\right)+\left(\nabla^{//}\mathbf{w}\right):\mathbf{c}:\frac{1}{2}\left((\mathbf{n}\partial_n\mathbf{s})+(\mathbf{n}\partial_n\mathbf{s})^T\right)]d^2\mathbf{r} + \int_\Gamma (\mathbf{nw}):\mathbf{c}:\frac{1}{2}\left(\left(\nabla^{//}\mathbf{s}\right)+\left(\nabla^{//}\mathbf{s}\right)^T\right)d\mathbf{r},$$

$$\int_V \rho(\partial_t + d(m))^2 \mathbf{s^4}\cdot\mathbf{w}d^2\mathbf{r} = -\int_V \left(\nabla^{//}\mathbf{w}\right):\mathbf{c}:\frac{1}{2}\left(\left(\nabla^{//}\mathbf{s}\right)+\left(\nabla^{//}\mathbf{s}\right)^T\right)d^2\mathbf{r},$$

$$\int_V \rho(\partial_t + d(m))^3 \mathbf{s^5}\cdot\mathbf{w}d^2\mathbf{r} = -\int_V p^{(m/n)}d'(n)\mathbf{nw}:\mathbf{c}:\frac{1}{2}\left(\left(\nabla^{//}\mathbf{s}\right)+\left(\nabla^{//}\mathbf{s}\right)^T\right)d^2\mathbf{r}, \quad (28)$$

where $d(m) = p^{(m/n)}d(n)$. Γ denotes the boundary of the PML that is not in contact with the regular region. Actually, we note that these second-order M-PML formulations could be written for different second-order numerical schemes, e.g. the SEM, FEM and FDM.

Due to the split components in the second-order wave equation are not linearly independent but coupling, there is no appropriate treatment of the corner regions (see Fig. 2) in the classical second-order PML formulations at present. Straightforwardly to add the damping variables in the two directions of the plane as in the first-order equations is no longer applicable. In order to handle the damping in the corner regions for the second-order classical PML, we have to reconstruct the PML formulations for damping the wavefields in different directions at the same time. The damping variables of the coupling terms in the equations face complicate mathematical procedure, while in the proposed second-order M-PML formulations (Eq. (28)), such difficulty is overcome in the uniform formulations.

The treatment of the corner regions for the proposed second-order M-PML is similar to that of the first-order M-PML (Meza-Fajardo and Papageogiou, 2008). Moreover, in corner regions, the formulations of Eq. (28) are replaced by seven independent equations (see Appendix).

5. Numerical validation

To illustrate the efficiency and the stability of the second-order M-PML formulations, we consider four numerical experiments of the elastic wave propagation with P-SV polarization in two-dimensional models with free surface. The wave equations in the regular region and the M-PML region are solved with the spectral element method (SEM). The accuracy of the proposed scheme is demonstrated by the comparison of the numerical and the analytical solutions in an isotropic half-space model. The results for the classical second-order PML are also presented for comparison. In the first two numerical experiments, we investigate the stability of the second-order M-PML for the isotropic half-space models. A high Poisson's ratio, which may yield unstable issue in the implementation of some ABCs, especially for Rayleigh wave modeling (Zeng et al., 2011), is considered. In the other two experiments, we simulate wave propagation in the anisotropic media, which may cause severely unstable issues.

For Case 1 of the isotropic half-space models, the size of the model is 192 m × 192 m. The source locates at (62.4 m, 0 m) on the free surface. The Ricker wavelet function is selected for the source time function with the central frequency of 50 Hz, and the time shift of 0.04 s. The PML absorbing boundaries are set in vertical and bottom edges (Fig. 2) with the span of 38.4 m. The model is discretized in 40 × 40 spectral elements with polynomial degree of $N = 4$. The physic parameters of the model are listed in Table 1.

For Case 2 of the isotropic half-space models, the size of the model is 55 m × 40 m. The source locates at (18.0 m, 0 m) on the free surface. The Ricker wavelet function is also selected with the central frequency and the time shift of 40 Hz and 0.04 s. The PML absorbing boundaries are set in vertical and bottom edges (Fig. 2) with the span of 8 m. The model is discretized in 55 × 40 spectral elements. The polynomial degree $N = 4$ is chosen in each element. The physic parameters of the model are listed in Table 1.

For Case 3 of the anisotropic half-space models, the size of the model is 616 m × 448 m. The Ricker wavelet source locates at (201.6 m, 0 m) on the free surface, with the central frequency of 22.5 Hz, and the time shift of 0.044 s. The PML absorbing boundaries are set 67.2 m in width around the three edges (see Fig. 2). The model consists of 55 × 40 spectral elements with polynomial degree of $N = 4$. The physic parameters of the model are listed in Table 2.

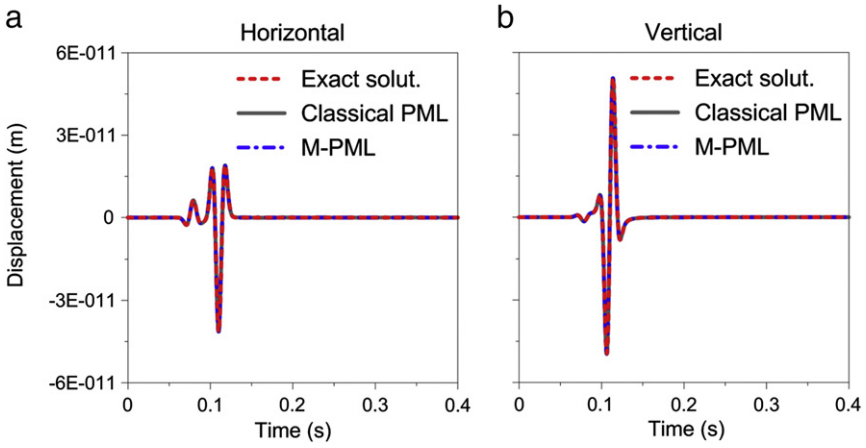


Fig. 3. Simulated seismograms of the horizontal (a) and the vertical (b) displacements on the surface with offset of 74.4 m in an elastic half-space model (Table 1, Case 1). The solid line, the dot-dashed line and the dashed line indicate the records obtained by the numerical simulations using the second-order classical PML (Komatitsch and Tromp, 2003), the M-PML and the analytical solution (Berg et al., 1994), respectively. The results show a good agreement. There is almost no spurious wave reflected off the boundary in the two numerical solutions.

For Case 4 of the anisotropic half-space models, the size of the model is 660 m × 480 m. The Ricker wavelet source locates at (216.0 m, 0 m) on the free surface, with the central frequency and the time shift of 40 Hz and 0.05 s. The PML absorbing boundaries are set 96 m in width around the three edges (see Fig. 2). The model is discretized in 55 × 40 spectral elements. The polynomial degree $N = 4$ is chosen in each element. The physic parameters of the model are listed in Table 2.

The boundaries between the regular region and the PML region are marked by the solid lines in the snapshots.

The damping coefficient can be written by (Komatitsch and Tromp, 2003)

$$d(x) = \frac{3\alpha}{2\delta}\log\left(\frac{1}{R}\right)\left(\frac{x}{\delta}\right)^2, \tag{29}$$

where δ is the width of the PML region, x is the distance between the calculating point to the interface of the PML and the regular region in

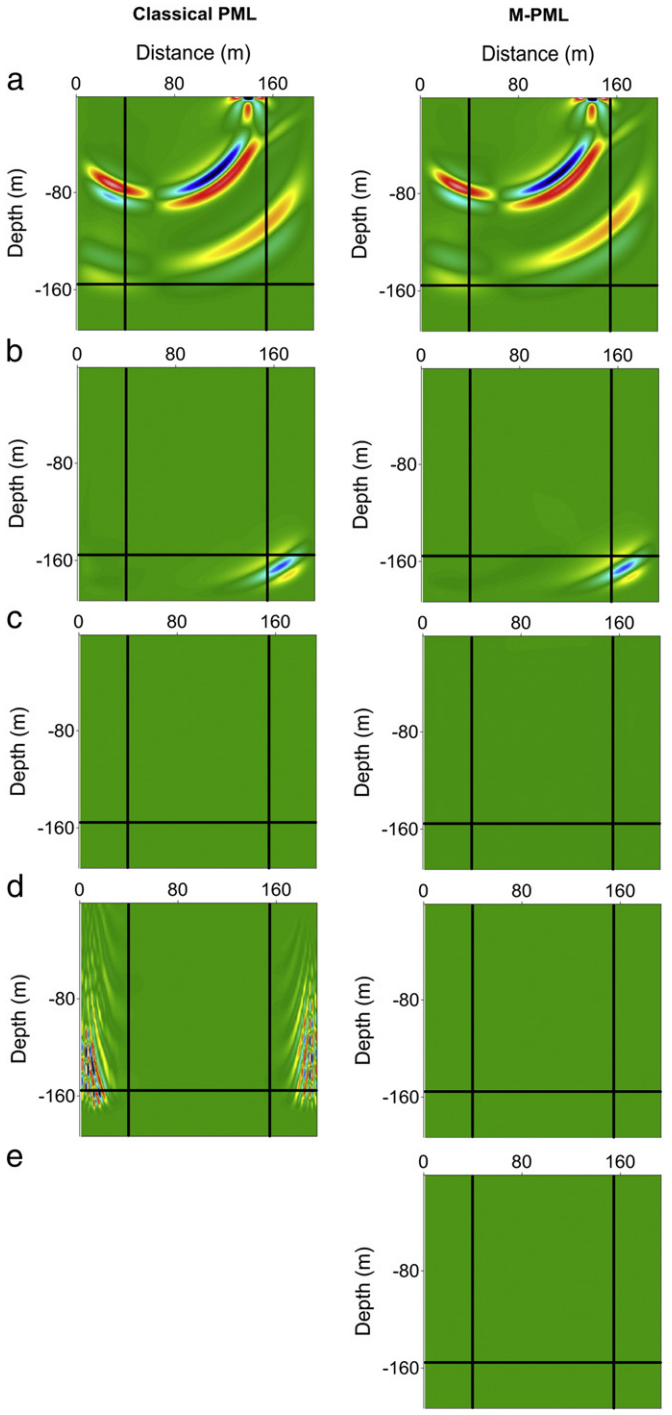


Fig. 4. Snapshots of the horizontal displacement simulated by the SEM in an isotropic elastic medium (Table 1, Case 1) at (a) $t = 0.112$ s, (b) $t = 0.21$ s, (c) $t = 0.35$ s, (d) $t = 5.6$ s, (e) $t = 6.8$ s. The left and right panels correspond to the second-order classical PML and the second-order M-PML ($p^{(x/z)} = p^{(z/x)} = 0.1$) terminations, respectively. No instability is observed in the M-PML termination.

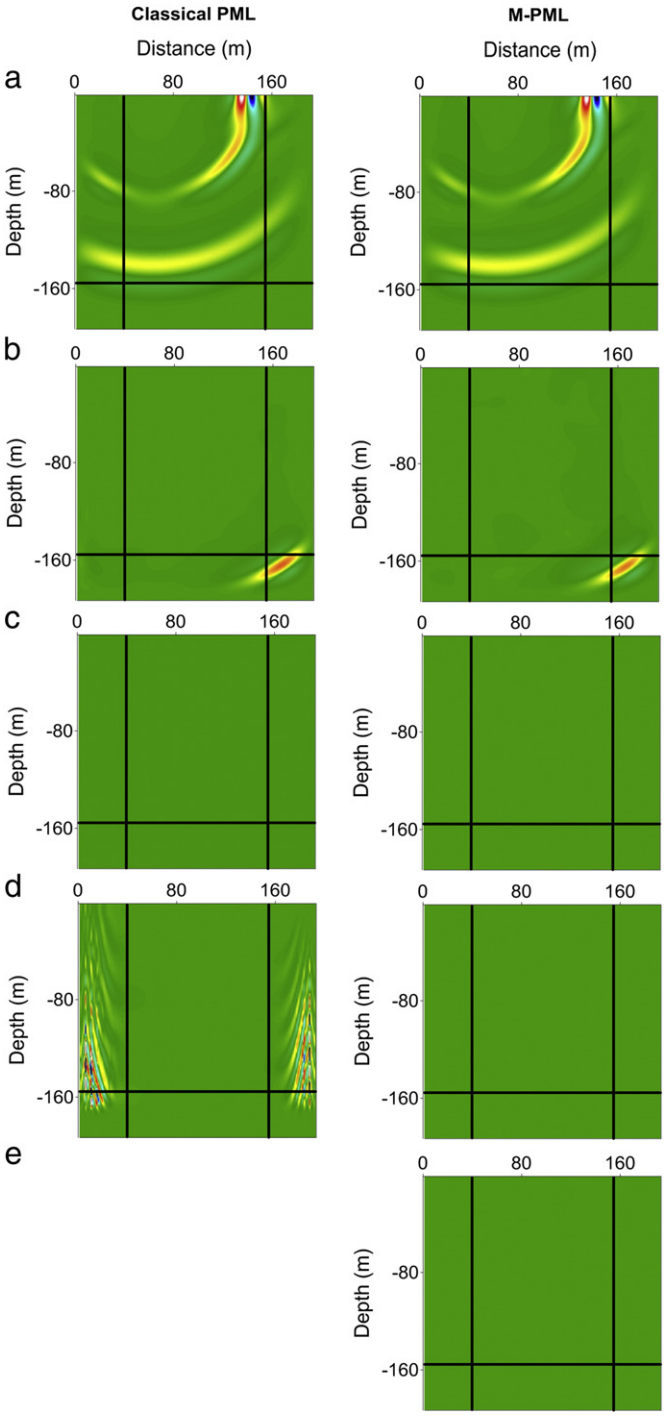


Fig. 5. Snapshots of the vertical displacement simulated by the SEM in an isotropic elastic medium (Table 1, Case 1) at (a) $t = 0.112$ s, (b) $t = 0.21$ s, (c) $t = 0.35$ s, (d) $t = 5.6$ s, (e) $t = 6.8$ s The left and right panels correspond to the second-order classical PML and the second-order M-PML ($p^{(x/z)} = p^{(z/x)} = 0.1$) terminations, respectively. The late-time instability is visible in the classical PML. No instability is observed in the M-PML termination.

the damping direction. R denotes the theoretical reflection coefficient after discretization, which can be chosen very small (a typical value as 8×10^{-4}).

Finally, in order to illustrate the efficiency of the second-order M-PML method, the total energy inside the regular region during the simulation is investigated. Thus, the total energy at each time can be expressed as (Matzen, 2011)

$$E = \frac{1}{2}\dot{\mathbf{s}}^T\mathbf{M}\dot{\mathbf{s}} + \frac{1}{2}\mathbf{s}^T\mathbf{K}\mathbf{s}, \tag{30}$$

where **M** is the mass matrix, **K** is the stiffness matrix, and **s** is the displacement vector.

Case 1. By using the second-order PML, the seismograms of the horizontal (Fig. 3a) and the vertical (Fig. 3b) displacements are recorded on the free surface with the offset of 74.4 m. The modeling seismograms (up to 0.4 s), both for the classical second-order PML and the proposed second-order M-PML, fit the analytical solution perfectly (Berg et al., 1994). No obvious reflectors are observed in both the horizontal and the vertical displacements.

The horizontal (Fig. 4) and the vertical (Fig. 5) displacement snapshots are obtained for the second-order classical PML and the second-order M-PML, respectively. The dominant Rayleigh waves can be easily identified (Figs. 4a, 5a ($t = 0.112$ s)). At $t = 0.21$ s, all the waves propagate outside the regular region. The damping of both the classical PML and the M-PML works perfectly inside the regular region. No obvious reflected wavefields are observed (Figs. 4b, 5b). When the waves propagate outside the finite simulated model, no visible wavefields can be identified, but only very low amplitude noise from the numerical procedures (Figs. 4c, 5c). For the long-time simulation ($t = 5.6$ s), the instability of the classical second-order PML appears (left panels of Figs. 4d, 5d). If the simulation is running for longer time, the values of unexpected wavefields will grow to cause instability and even overflow. However, no instability is identified for the M-PML (right panels of Figs. 4d–e, 5d–e), which can also be verified in the energy curves (Fig. 6).

Dmitriev and Lisitsa (2011) have proved that the M-PML method is not perfectly matched, and the reflectivity of the M-PML exceeds that of the classical PML at some time steps. We can observe these phenomena in the energy curves around 0.3 s to 0.5 s (Fig. 6). These slight reflections are insignificant and acceptable in the simulation. The total energy decays continuously for the M-PML, however, after around 2 s, the energy increases illogically for the classical PML, which results in the instability.

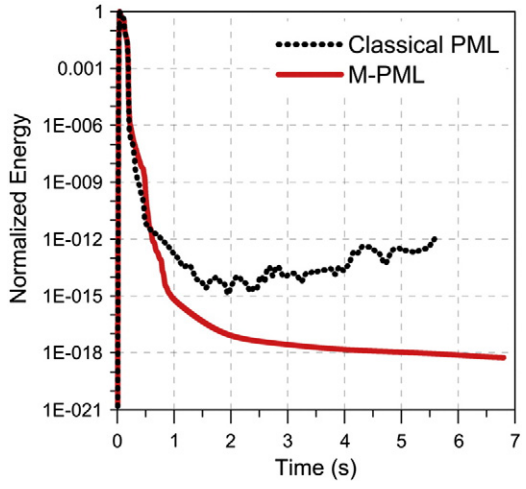


Fig. 6. Energy curves versus time for the classical PML and the M-PML in the regular region for Case 1.

Case 2. In this section, the efficiency of the second-order M-PML is validated in an isotropic medium with a high Poisson's ratio of $v = 0.47$, which may yield unstable issue in the implementation of some ABCs (Clayton and Engquist, 1977; Mahrer, 1990; Stacey, 1988), especially for Rayleigh wave modeling (Zeng et al., 2011).

For the P-wave velocity greatly larger than S-wave velocity, when the P-wave is far away the source, the S-wave propagates nearby the source (Figs. 7a, 8a). At $t = 0.2$ s, the body and surface waves have propagated into the PML region, and are efficiently absorbed (Figs. 7b, 8b). For the classical PML, when the body and surface waves propagate outside the finite simulated model, most of the spurious reflections are suppressed (Figs. 7c, 8c) except for some small unexpected energies appear nearby the corner regions (left panels of Figs. 7c, 8c). These unexpected energies aggravate as time goes by (left panels of Figs. 7d, 8d). At $t = 1.6$ s the instable energies erode the regular region and the run breaks down. In contrary to

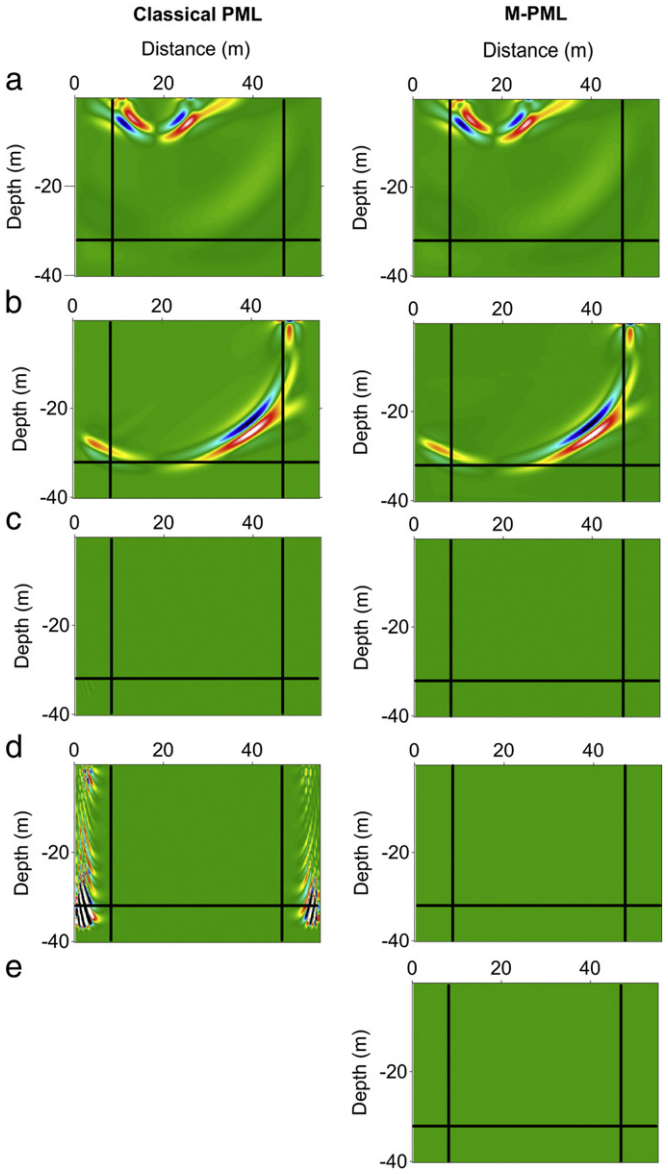


Fig. 7. Snapshots of the horizontal displacement simulated by the SEM in an isotropic elastic medium with high Poisson's ratio (Table 1, Case 2) at (a) $t = 0.08$ s, (b) $t = 0.2$ s, (c) $t = 0.56$ s, (d) $t = 1.04$ s, (e) $t = 7.9$ s. The left and right panels correspond to the second-order classical PML and the second-order M-PML ($p^{(x/z)} = p^{(z/x)} = 0.6$) terminations, respectively. No instability is observed in the M-PML termination.

the classical PML, the proposed M-PML has very good performance to suppress the spurious reflections in whole simulation time (right panels of Figs. 7c–e, 8c–e).

The energy curves in the regular region for the second-order classical PML and the second-order M-PML display in Fig. 9. The insignificant and acceptable reflections of the M-PML, that exceed that of the classical PML, are observed in the energy curves around 0.3 s to 1.0 s. The simulation is performed up to 7.9 s, and no instability represented by the energy increment is observed for the M-PML. The total energy for the M-PML decays smoothly and steadily, which demonstrates the numerical stability for the M-PML, even for high Poisson's ratio. However, after around 1.5 s, the energy increases illogically for the classical PML, which results in the instability. The simulation cannot be normally performed after that time step.

Case 3. This particular orthotropic anisotropic model has been studied previously (The normalized parameters are given in Bécache et al., 2001). Meza-Fajardo and Papageogiou (2008) designed a stable M-PML for this particular medium in the first-order equations. The designed half-space model with the free surface undoubtedly augments the instabilities. Observing the snapshots of the horizontal and the vertical displacements for the classical PML and the M-PML (Figs. 10 and 11), the wavefields in the regular region spread normally (Figs. 10a, 11a), while the body waves are splitting. The instability appears at about $t = 0.32$ s for the classical PML (left panels of Fig. 10b, 11b), at that time the split slow waves propagate into the absorbing region. This instability becomes visible and the wavefields are severely polluted in the regular region before $t = 0.56$ s (left panels of Figs. 10c, 11c), while the propagating wavefields have not totally been out of the regular region. We can identify these phenomena in the energy curves for the classical PML (Fig. 12). The spurious energies erode the whole regular region at $t = 0.6$ s and the run breaks down (Fig. 12). On the contrary, no instability is observed during the simulated procedure with time up to 2 s for the proposed second-order M-PML (right panels of Figs. 10, 11). Some slight reflections can be observed at $t = 0.56$ s (right panels of Figs. 10c, 11c), which may be caused by the imperfect match for the M-PML. Nevertheless, the total energy in the regular region for the M-PML damps smoothly and steadily. No instability appears by illogical energy increasing for the M-PML in whole simulation time (up to 2 s, see Fig. 12).

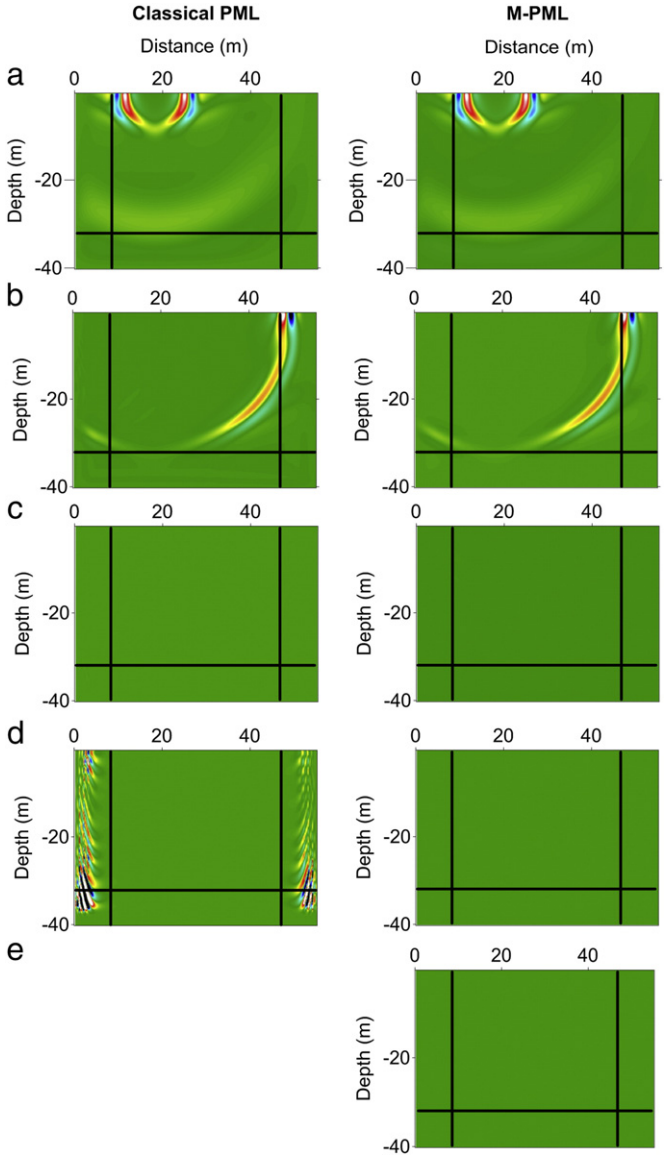


Fig. 8. Snapshots of the vertical displacement simulated by the SEM in an isotropic elastic medium with high Poisson's ratio (Table 1, Case 2) at (a) $t = 0.08$ s, (b) $t = 0.2$ s, (c) $t = 0.56$ s, (d) $t = 1.04$ s, (e) $t = 7.9$ s. The left and right panels correspond to the second-order classical PML and the second-order M-PML ($p^{(x/z)} = p^{(z/x)} = 0.6$) terminations, respectively. The late-time instability is visible in the classical PML. No instability is observed in the M-PML termination.

Case 4. The classical PML and the C-PML may suffer from the late-time instabilities for the numerical simulation in anisotropic media (Matzen, 2011; Meza-Fajardo and Papageogiou, 2008) involving the model of common crystals like the metal, zinc (Komatitsch and Martin, 2007). Although the first-order M-PML scheme can be applied to handle the situation (Meza-Fajardo and Papageogiou, 2008), the second-order approach need to be further investigated in this specific model.

In view of the snapshots of the horizontal and the vertical displacements (Figs. 13 and 14), the split body waves are well absorbed by the two PML methods (Figs. 13a–d, 14a–d). At $t = 1.4$ s (left panels of Figs. 13e and 14e), the numerical noise derived by the instability has polluted the regular region for the classical PML. The unexpected wavefields grow so severely in the vertical strips that the calculation cannot be continued any more.

On the other hand, no instability is observed in the snapshots for the proposed second-order M-PML (right panels of Figs. 13 and 14). The energy curve for the M-PML also demonstrates its superior performance (Fig. 15). In the long-time simulation, there is no instability appearing in the regular region. The total energy for the M-PML decays smoothly and steadily during simulation time. Although it is less than that for the M-PML around 0.3 s to 0.8 s, the total energy for the classical PML

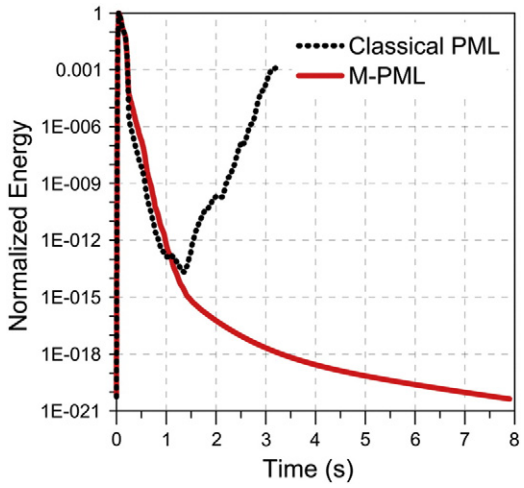


Fig. 9. Energy curves versus time for the classical PML and the M-PML in the regular region for Case 2.

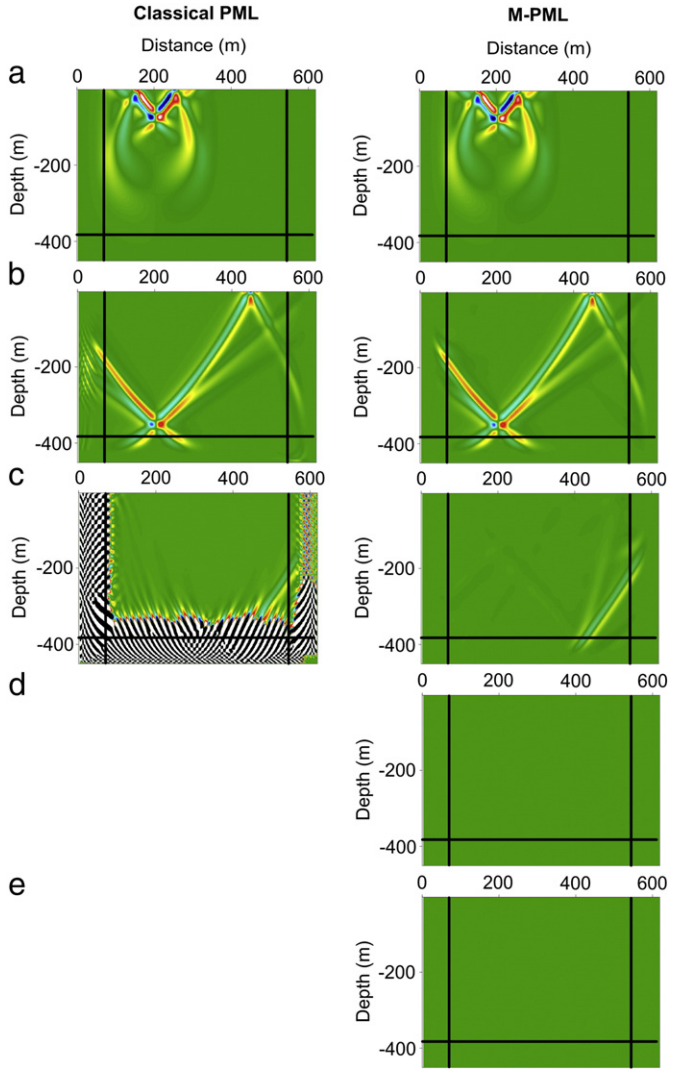


Fig. 10. Snapshots of the horizontal displacement simulated by the SEM in an orthotropic anisotropic elastic medium (Table 2, Case 3) at (a) $t = 0.1$ s, (b) $t = 0.32$ s, (c) $t = 0.56$ s, (d) $t = 1.04$ s, (e) $t = 2$ s. The left and right panels correspond to the second-order classical PML and the second-order M-PML ($p^{(x/z)} = 0.25$, $p^{(z/x)} = 0.3$) terminations, respectively. No instability is observed for the M-PML termination.

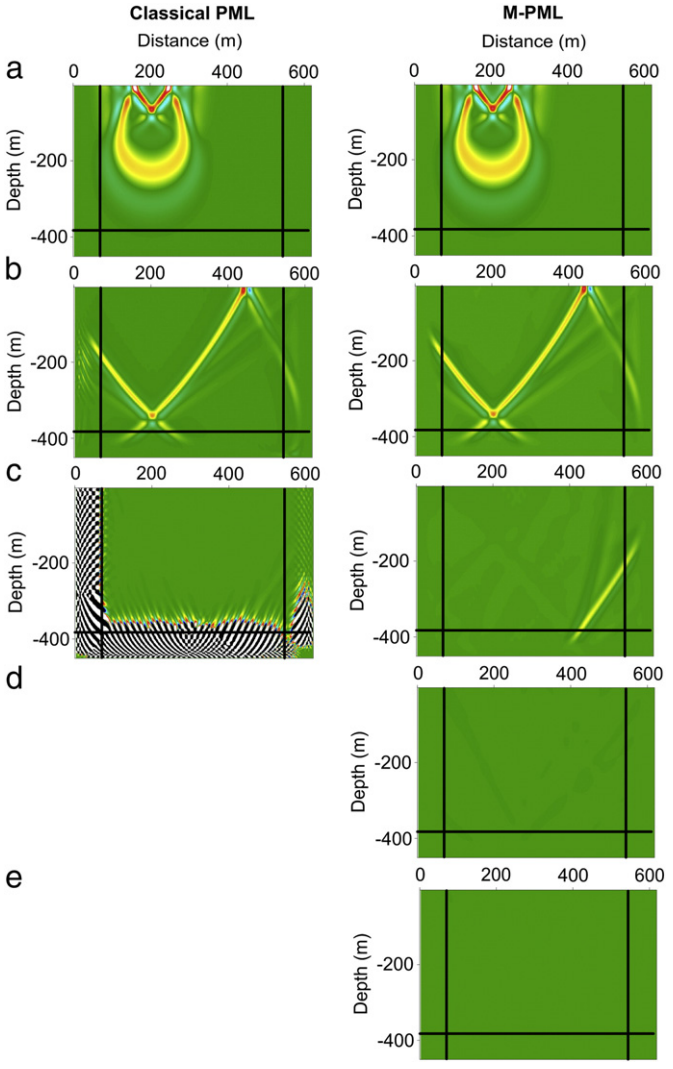


Fig. 11. Snapshots of the vertical displacement simulated by the SEM in an orthotropic anisotropic elastic medium (Table 2, Case 3) at (a) $t = 0.1$ s, (b) $t = 0.32$ s, (c) $t = 0.56$ s, (d) $t = 1.04$ s, (e) $t = 2$ s. The left and right panels correspond to the second-order classical PML and the second-order M-PML ($p^{(x/z)} = 0.25$, $p^{(z/x)} = 0.3$) terminations, respectively. The instability appears when the split slow waves propagate into the absorbing region. No instability is observed for the M-PML termination.

increases illogically after 0.8 s (Fig. 15), which results in the instability. The instability aggravates as time goes by and the run will break down.

6. Conclusions

A nonconvolutional perfectly matched layer, referred to as the multiaxial perfectly matched layer (M-PML) for the second-order elastic wave equations, has been proposed and numerically validated. The primary of this work is to show that an efficient M-PML implementation is indeed possible when e.g. the existing displacement-based (second-order) SEM or FEM codes need to tackle the limited region elastic wave simulation problems. It is verified that the absorbing boundary represented in the second-order wave equations can be well constructed by the M-PML formulations. The present scheme is stable and efficient, and allows us to simulate the elastic wavefields, especially in anisotropic media for long time using the spatial second-order wave equations in the SEM, FEM and FDM.

Through the comparison with the second-order classical PML proposed by Komatitsch and Tromp (2003), better stability is achieved by the second-order M-PML formulations in the several numerical

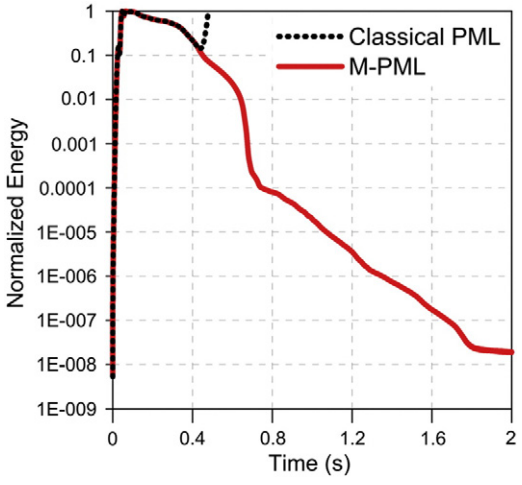


Fig. 12. Energy curves versus time for the classical PML and the M-PML in the regular region for Case 3.

experiments. In the long-time simulation, the proposed second-order M-PML scheme keeps stable, even for the media with high Poisson's ratio (Case 2), and the specific orthotropic anisotropy (Cases 3, 4) which may usually yield unstable issue in the implementation of the PML methods, such as the first-order C-PML. The resulting elastodynamics remain valid for the wave propagation of different frequencies and directions, and show advantages to the second-order classical PML.

The established formulations are similar to the M-PML in the first-order wave equations. The generality of the proposed second-order M-PML scheme leads to its adaptability in other second-order formulations, as in acoustics and electromagnetics. Extension of the proposed scheme to three-dimensional problems is straightforward and will be studied in the future.

Acknowledgments

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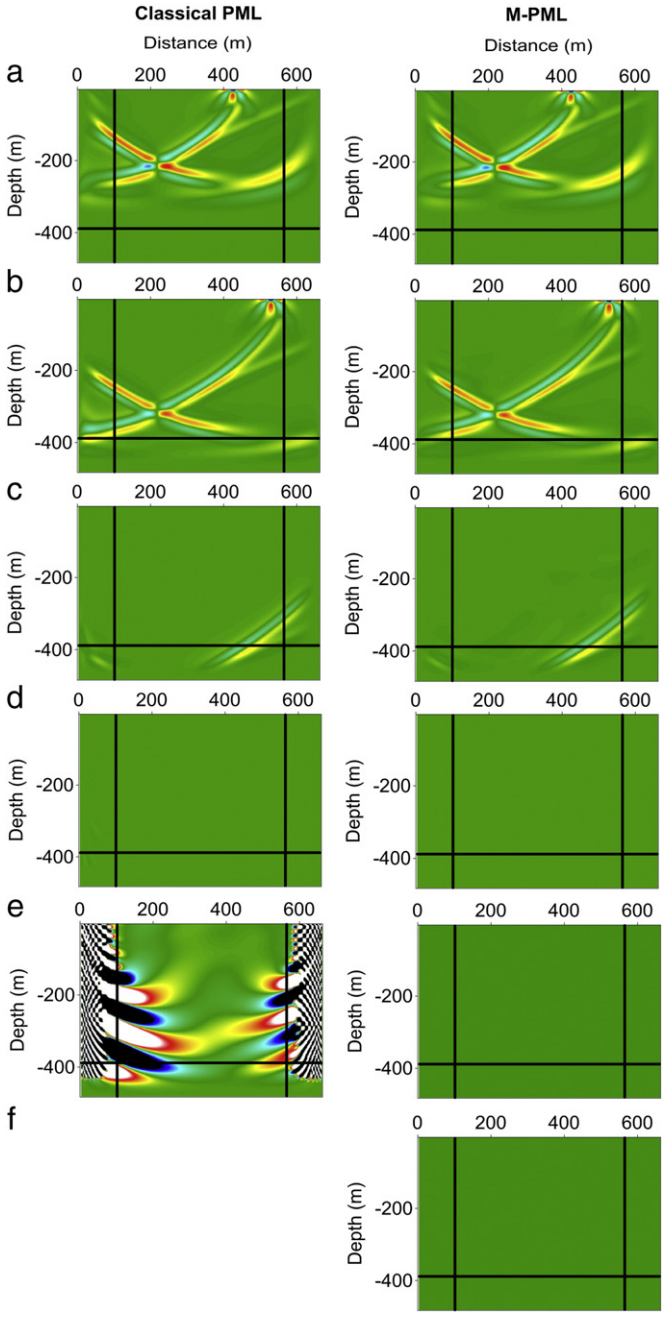


Fig. 13. Snapshots of the horizontal displacement simulated by the SEM in an orthotropic anisotropic elastic medium (Table 2, Case 4) at (a) $t = 0.15$ s, (b) $t = 0.2$ s, (c) $t = 0.3$ s, (d) $t = 0.5$ s, (e) $t = 1.4$ s, (f) $t = 5.4$ s. The left and right panels correspond to the second-order classical PML and the second-order M-PML ($p^{(x/z)} = p^{(z/x)} = 0.15$) terminations, respectively. No instability is observed for the M-PML termination.

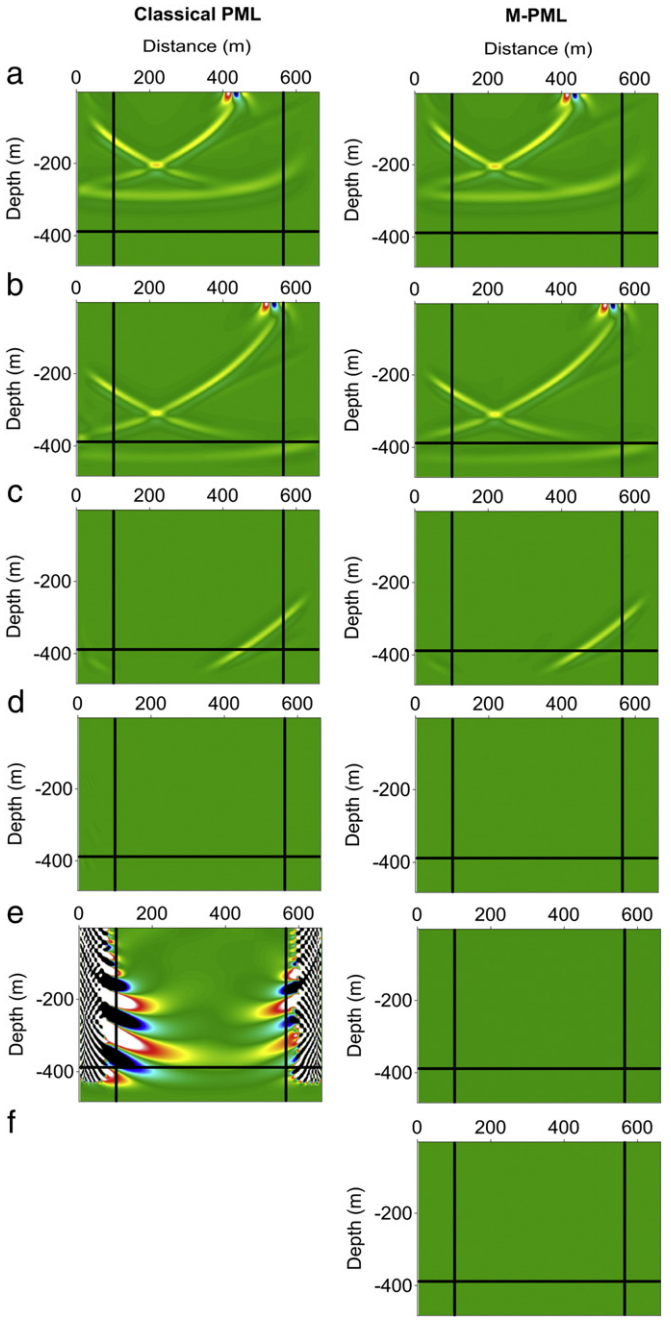


Fig. 14. Snapshots of the vertical displacement simulated by the SEM in an orthotropic anisotropic elastic medium (Table 2, Case 4) at (a) $t = 0.15$ s, (b) $t = 0.2$ s, (c) $t = 0.3$ s, (d) $t = 0.5$ s, (e) $t = 1.4$ s, (f) $t = 5.4$ s. The left and right panels correspond to the second-order classical PML and the second-order M-PML ($p^{(x/z)} = p^{(z/x)} = 0.15$) terminations, respectively. The late-time instability is visible in the classical PML. No instability is observed in the M-PML termination.

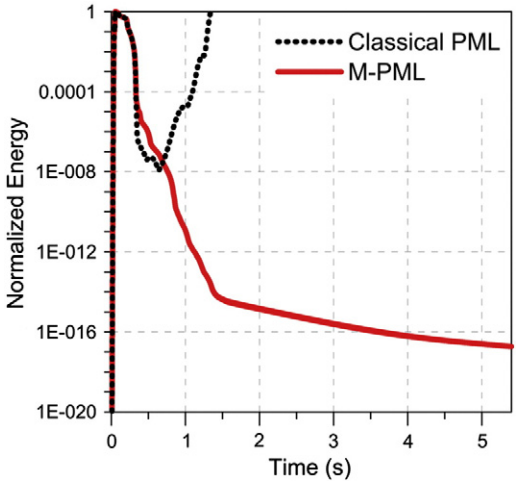


Fig. 15. Energy curves versus time for the classical PML and the M-PML in the regular region for Case 4.

Appendix

The seven independent equations in the corner region can be written by

$$\int_V \rho\left(\partial_t + d(n) + p^{n/m}d(m)\right)^2 \mathbf{s^1}\cdot\mathbf{w}d^2\mathbf{r} = -\int_V (\mathbf{n}\partial_n\mathbf{w}):\mathbf{c}:\frac{1}{2}\left((\mathbf{n}\partial_n\mathbf{s})+(\mathbf{n}\partial_n\mathbf{s})^T\right)d^2\mathbf{r}$$
$$+\int_\Gamma (\mathbf{nw}):\mathbf{c}:\frac{1}{2}\left((\mathbf{n}\partial_n\mathbf{s})+(\mathbf{n}\partial_n\mathbf{s})^T\right)d\mathbf{r},$$
$$\int_V \rho\left(\partial_t + d(n) + p^{n/m}d(m)\right)^3 \mathbf{s^2}\cdot\mathbf{w}d^2\mathbf{r} = -\int_V d'(n)\mathbf{nw}:\mathbf{c}:\frac{1}{2}\left((\mathbf{n}\partial_n\mathbf{s})+(\mathbf{n}\partial_n\mathbf{s})^T\right)d^2\mathbf{r},$$
$$\int_V \rho[\partial_t{}^2 + \left(d(m)+p^{(m/n)}d(n)+d(n)+p^{(n/m)}d(m)\right)\partial_t$$
$$+\left(d(m)+p^{(m/n)}d(n)\right)\left(d(n)+p^{(n/m)}d(m)\right)]\mathbf{s^3}\cdot\mathbf{w}d^2\mathbf{r}$$
$$= -\int_V [(\mathbf{n}\partial_n\mathbf{w}):\mathbf{c}:\frac{1}{2}\left(\left(\nabla^{//}\mathbf{s}\right)+\left(\nabla^{//}\mathbf{s}\right)^T\right)$$
$$+\left(\nabla^{//}\mathbf{w}\right):\mathbf{c}:\frac{1}{2}\left((\mathbf{n}\partial_n\mathbf{s})+(\mathbf{n}\partial_n\mathbf{s})^T\right)]d^2\mathbf{r}+\int_\Gamma (\mathbf{nw}):\mathbf{c}:\frac{1}{2}\left(\left(\nabla^{//}\mathbf{s}\right)+\left(\nabla^{//}\mathbf{s}\right)^T\right)d\mathbf{r},$$
$$\int_V \rho\left(\partial_t + d(m) + p^{(m/n)}d(n)\right)^2 \mathbf{s^4}\cdot\mathbf{w}d^2\mathbf{r} = -\int_V \left(\nabla^{//}\mathbf{w}\right):\mathbf{c}:\frac{1}{2}\left(\left(\nabla^{//}\mathbf{s}\right)+\left(\nabla^{//}\mathbf{s}\right)^T\right)d^2\mathbf{r},$$
$$\int_V \rho\left(\partial_t + d(m) + p^{(m/n)}d(n)\right)^2 \left(\partial_t + d(n) + p^{(n/m)}d(m)\right)\mathbf{s^5}\cdot\mathbf{w}d^2\mathbf{r}$$
$$= -\int_V p^{(m/n)}d'(n)\mathbf{nw}:\mathbf{c}:\frac{1}{2}\left(\left(\nabla^{//}\mathbf{s}\right)+\left(\nabla^{//}\mathbf{s}\right)^T\right)d^2\mathbf{r},$$
$$\int_V \rho\left(\partial_t + d(n) + p^{(n/m)}d(m)\right)^2 \left(\partial_t + d(m) + p^{(m/n)}d(n)\right)\mathbf{s^6}\cdot\mathbf{w}d^2\mathbf{r}$$
$$= -\int_V p^{(n/m)}d'(m)\mathbf{mw}:\mathbf{c}:\frac{1}{2}\left((\mathbf{n}\partial_n\mathbf{s})+(\mathbf{n}\partial_n\mathbf{s})^T\right)d^2\mathbf{r},$$
$$\int_V \rho\left(\partial_t + d(m) + p^{(m/n)}d(n)\right)^3 \mathbf{s^7}\cdot\mathbf{w}d^2\mathbf{r} = -\int_V d'(m)\mathbf{mw}:\mathbf{c}:\frac{1}{2}\left(\left(\nabla^{//}\mathbf{s}\right)+\left(\nabla^{//}\mathbf{s}\right)^T\right)d^2\mathbf{r}, \quad (29)$$

where $\mathbf{s}^1$, $\mathbf{s}^2$, $\mathbf{s}^3$, $\mathbf{s}^4$, $\mathbf{s}^5$, $\mathbf{s}^6$ and $\mathbf{s}^7$ represent the real displacements with different direction couplings.

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